

ESHAP chemotherapy

ESHAP is a cancer drug treatment. It is used to treat blood cancers such as Hodgkin lymphoma, non-Hodgkin lymphoma and myeloma.

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What is ESHAP?

ESHAP is used to treat [blood cancers](#) such as [Hodgkin lymphoma](#), [non-Hodgkin lymphoma](#) and [myeloma](#). It may be given before a stem cell transplant. It is best to read this information with our information about [chemotherapy](#), [stem cell transplants](#) and the [type of cancer](#) you have.

ESHAP is a combination of chemotherapy and a steroid drug:

- [etoposide](#)
- [cytarabine](#)
- [cisplatin](#)
- methylprednisolone (a [steroid](#)).

Sometimes a drug called rituximab is given with ESHAP. This is called R-ESHAP. We have separate information about [rituximab](#).

Your doctor will talk to you about this treatment and its possible side effects before you agree ([consent](#)) to have treatment.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

How ESHAP is given

You usually have ESHAP during a stay in hospital. Some people might have ESHAP as an outpatient. A chemotherapy nurse will give it to you.

During your course of treatment, you will meet someone from your cancer or haematology team such as a:

- blood cancer doctor (haematologist) and sometimes a cancer doctor (oncologist)
- chemotherapy nurse or a specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

Before or on the day of treatment, you will have a blood test. This is to check that it is safe for you to have treatment.

You will meet with a doctor, nurse or pharmacist before you have treatment. They will talk to you about your blood results and ask how you have been feeling. If your blood results are okay, the pharmacy team will prepare your cancer drugs.

Your nurse will usually give you anti-sickness drugs before the cancer drugs. You will also be given extra fluids through a drip before and after chemotherapy. This is to protect your kidneys.

You will have your cancer treatment through 1 of the following:

- a [central line](#) – a fine tube that goes under the skin of your chest and into a vein close by
- a [PICC line](#) – a fine tube that is put into a vein in your arm and goes up into a vein in your chest.

The drugs run through a pump, which gives you the treatment over a set time:

- cisplatin is usually given over 22-24 hours.
- cytarabine over 2-3 hours
- etoposide over 1 hour
- methylprednisolone over 15-30 minutes.

Sometimes methylprednisolone is given as tablets instead of a drip.

Your cancer team may also give you anti-sickness drugs and other medicines to take home. Take all your capsules or tablets exactly as they tell you to.

If you have ESHAP as an outpatient, you have the cisplatin infusions through a small portable pump that you can take home. Your nurse or pharmacist will explain how to look after the pump.

You need to return to the hospital each day for the etoposide and methylprednisolone, and to have the cisplatin infusion changed.

Your course of chemotherapy

You usually have a course of several cycles of treatment over a few months. Your cancer team will discuss your treatment plan with you. They may give you a copy of the plan to take home.

Depending on your blood test results each cycle of ESHAP is either 21 or 28 days. You have treatment on days 1-5 of each cycle. This is followed by a rest period for either 16 or 23 days, until the next cycle begins.

You have the drugs on the following days:

- day 1 - cisplatin, cytarabine and etoposide and methylprednisolone
- days 2, 3 and 4 - cisplatin, etoposide and methylprednisolone
- day 5 - the cisplatin from day 4 finishes.

Sometimes cytarabine may be given on either day 5 or 6 instead of day 1.

Your cancer team will talk to you about how many cycles you will have and discuss your treatment plan with you.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Side effects while treatment is being given

Some people may have side effects while they are having a cancer drug or shortly after they have it:

Allergic reaction

Some people have an allergic reaction while having this treatment. Signs of a reaction can include:

- feeling hot or flushed
- shivering
- itching
- a skin rash
- feeling dizzy or sick
- a headache
- feeling breathless or wheezy
- swelling of your face or mouth
- pain in your back, tummy or chest.

Your nurse will check you for signs of a reaction during your treatment. If you feel unwell or have any of these signs, tell them straight away. If you do have a reaction, they can treat it quickly.

Sometimes a reaction happens a few hours after treatment. If you develop any of these signs or feel unwell after you get home, contact the hospital straight away on the 24-hour number.

Always call 999 if swelling happens suddenly or you are struggling to breathe.

Very common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this treatment.

Cytarabine syndrome

Cytarabine may cause some symptoms 6 to 12 hours after it has been given. This is called cytarabine syndrome. Signs can include:

- a high temperature or chills
- a rash
- pain in the eyes, bones, tummy or chest.

You may be given medicines before treatment to help prevent or reduce these symptoms.

If you have any of these symptoms, tell your nurse or contact the hospital straight away.

Risk of infection

This treatment can reduce the number of white blood cells in your blood. These cells fight infection. If your white blood cell count is low, you may be more likely to get an [infection](#). A low white blood cell count is called neutropenia.

An infection can be very serious when the number of white blood cells is low. Your doctor may give you antibiotics and other drugs to take during this treatment. These may help reduce your risk of getting an infection.

But it is still important to get any infection treated as soon as possible. If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature
- you have symptoms of an infection.

Symptoms of an infection include:

- feeling shivery and shaking
- a sore throat
- a cough

- breathlessness
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

It is important to follow any specific advice your cancer treatment team gives you.

Your white blood cell count will usually return to normal before your next treatment. You will have a blood test before having more treatment. If your white blood cell count is low, your doctor may delay your treatment for a short time, until your cell count increases.

You may be given a drug called [G-CSF](#). This encourages the body to make more white blood cells. You have it as a small injection under the skin.

Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds
- bleeding gums
- heavy periods
- blood in your urine (pee) or stools (poo)
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

Feeling sick

The nurses will give you anti-sickness drugs regularly. This will help prevent or control sickness during your treatment in hospital. [If you feel sick](#) or are sick (vomit), tell your nurse or doctor straight away. They can change the anti-sickness treatment. Or they might change the way you have it to something that works better for you.

Feeling tired (fatigue)

This treatment can make you [feel very tired](#) and you will need a lot of rest. You may get tired easily for some months after treatment ends. Gentle exercise, like short walks, can help you feel less tired. The tiredness will slowly get better.

If you feel sleepy, do not drive or use machinery.

Hair loss

You will lose all the hair from your head. You may also lose your eyelashes and eyebrows, as well as other body hair. [Hair loss](#) usually starts after your first or second treatment.

If you want to cover up hair loss, there are different ways you can do this. Your nurse will give you information about coping with hair loss.

Remember to protect your skin from the sun. Use suncream with a sun protection factor (SPF) of at least 30 on your scalp. Or cover up with a hat or scarf.

Hair loss is almost always temporary. Your hair will usually grow back after treatment ends.

Tummy (abdominal) pain

If you get pain or discomfort in your tummy (abdomen), tell your doctor or nurse. Your doctor may prescribe drugs to prevent this or to improve your symptoms. Tell your doctor if:

- the pain does not get better
- the pain becomes severe
- if you have blood in your vomit (sick) or stools (poo).

Constipation

This treatment can cause [constipation](#). Constipation means that you are not able to pass stools (poo) as often as you normally do. It can become difficult or painful. Here are some tips that may help:

- Drink at least 2 litres (3½ pints) of fluids each day.
- [Eat high-fibre foods](#), such as fruit, vegetables and wholemeal bread.
- Do regular gentle exercise, like going for short walks.

If you have constipation, contact the hospital on the 24-hour number for advice. They can give you drugs called laxatives to help.

If you have not been able to pass stools for over 2 days and are being sick, contact the 24-hour number straight away.

Loss of appetite

This treatment can affect your appetite. Don't worry if you do not eat much for 1 or 2 days. But if your appetite does not come back after a few days, or if you are losing weight, tell your doctor, nurse or pharmacist. They can give you advice. They may give you food or drink supplements. Or they may suggest changes to your diet or eating habits to help.

Effects on the liver

This treatment can affect how the liver works. You will have regular blood tests to check this. Sometimes liver changes can be serious. Contact the hospital straight away on the 24-hour number if you have any of these symptoms:

- yellow skin or eyes
- feeling very sleepy
- dark urine (pee)
- unexplained bleeding or bruising
- pain in the right side of your tummy (abdomen).

If you have black or brown skin, changes in your skin colour may be less noticeable. You might first notice the whites of your eyes look yellow, or that the colour of your urine has changed.

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Sore mouth and throat

This treatment may cause a [sore mouth and throat](#). You may also get mouth ulcers. This can make you more likely to get a mouth or throat infection. Use a soft toothbrush to clean your teeth or dentures in the morning, at night and after meals.

Contact the hospital straight away on the 24-hour number, if:

- a sore mouth or throat affects how much you can drink or eat
- your mouth, tongue, throat or lips have any blisters, ulcers or white patches.

They can give you advice, and mouthwash or medicines to help with the pain or to treat any infection. Follow their advice and make sure you:

- drink plenty of fluids
- avoid alcohol and tobacco
- avoid food or drinks that irritate your mouth and throat.

Diarrhoea

This treatment may cause [diarrhoea](#). Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. You may also have stomach cramps. If you have a stoma, it may be more active than usual.

If you are passing loose stools 3 or more times a day and this is not normal for you, contact the hospital as soon as possible on the 24-hour number. Follow the advice they give you about:

-

taking anti-diarrhoea medicines

- drinking enough fluids to keep you hydrated and to replace lost salts and minerals
- any changes to your diet that might help.

They might also ask you for a specimen of your stool to check for infection.

Effects on the kidneys

Cisplatin can affect how the kidneys work. You will have blood tests before and during treatment to check how well your kidneys are working.

Before and after each treatment, your nurse will give you extra fluids through a drip. This is to protect your kidneys.

Drinking fluids also helps protect your kidneys. The advice is usually to try to drink at least 2 litres (3½ pints) of fluid each day. But follow any advice from your doctor, nurse or pharmacist about how much is right for you.

Contact the hospital on the 24-hour number if you are:

- not able to drink as much as you have been asked to – for example, if you feel sick
- sick (vomit) or have diarrhoea
- passing less urine or peeing less often than usual.

Skin changes

Chemotherapy can affect your skin. If your skin feels dry, try using soap-free cleansers and unperfumed moisturising cream every day.

This treatment can also:

- cause a rash, which may be itchy
- make your skin darker in some areas
- make any area treated with radiotherapy become red or sore. If you have white skin the area will become red and if you have black or brown skin the area might become darker.
- make you more sensitive to the sun.

Your [skin may burn more easily](#) during treatment and for several months after. Use a sun cream of at least SPF 30. SPF stands for sun protection factor. Cover up with clothing and wear a hat.

Always tell your doctor, nurse or pharmacist about any [skin changes](#). They can give you advice or prescribe creams or medicines to help. Changes to your skin are usually temporary and improve when treatment ends.

Blood clot risk

Cancer and some cancer treatments can increase the risk of a [blood clot](#). Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
-

reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker

- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Hearing changes

This treatment may cause hearing changes, including hearing loss. You may have ringing in the ears. This is called tinnitus. You may also become unable to hear some high-pitched sounds. Hearing changes may get better after this treatment ends. But this does not always happen. If you notice any changes in your hearing, tell your doctor, nurse or pharmacist.

Effects on the eyes

Cytarabine may make your eyes feel sore, red, dry or itchy. This is called conjunctivitis. Your doctor may prescribe steroid eye drops to help prevent this. It is important to use these as you are told to.

Do not wear contact lenses if you are using eye drops or if your eyes are sore. Your doctor can tell you when you can wear them again.

Cytarabine may also:

- make your eyes more sensitive to light
- make your eye water
- cause blurry vision
- cause eye pain.

If you have pain or notice any change in your vision, always tell your doctor or nurse.

Raised blood sugar levels

This treatment can raise your blood sugar levels and increase the risk of diabetes.

Signs of raised blood sugar include:

- feeling thirsty
- needing to pass urine (pee) more often than usual
- feeling more tired than usual.

Tell your doctor or nurse if you have these symptoms. They can do tests to check your blood sugar level and give you advice.

If you already have [diabetes](#), your blood sugars may be higher than usual. You may need to check them more often, or make changes to your diabetic treatment. Your diabetic doctor or nurse will talk to you about how to manage this.

Mood and behaviour changes

Steroids can affect your mood and behaviour. They can cause:

- feelings of [anxiety](#) or restlessness
- mood swings (moods that go up and down)
- a low mood or [depression](#)
- difficulty sleeping.

If you notice any changes in your mood or behaviour, tell your doctor or nurse. They may make some changes to your treatment if the side effects are causing you problems.

Numb or tingling hands or feet (peripheral neuropathy)

This treatment may affect the nerves in your fingers and toes. This can cause numbness, tingling or pain in your hands or feet. This is called [peripheral neuropathy](#). You might find it hard to do fiddly tasks such as fastening buttons or tying shoelaces.

If you have these symptoms, always tell your doctor, nurse or pharmacist. They sometimes need to change the drug or the dose of the drug. The symptoms usually improve slowly after treatment ends. But for some people they continue and are a long-term side effect of treatment.

Tumour lysis syndrome (TLS)

Some people are at risk of developing a condition called tumour lysis syndrome (TLS). The risk is highest at the beginning of treatment. TLS happens when treatment makes large numbers of cancer cells die and break down quickly. This releases lots of waste products into the blood and can affect the kidneys and heart.

TLS can be prevented or treated. You will have regular blood tests to check for TLS. You may have:

- extra fluids through a drip
- medicines such as allopurinol tablets or [rasburicase](#) through a drip.

Drinking at least 2 litres (3½ pints) of fluid a day will also help.

Effects on the heart

This treatment can affect how the heart works. You may have [tests](#) to check how well your heart is working. These may be done before, during and after treatment.

If the treatment is causing [heart problems](#), your doctor may change the type of treatment you are having.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- breathlessness
-

- dizziness
- changes to your heartbeat (palpitations)
- swollen feet and ankles.

Other conditions can cause these symptoms, but it is important to get them checked by a doctor.

Always call **999** if you have:

- chest pain, pressure, heaviness, tightness or squeezing across the chest
- difficulty breathing.

Effects on the lungs

This treatment can cause inflammation of the lungs. This is called pneumonitis. Contact the hospital straight away on the 24-hour number if you notice any of these changes during treatment or after it ends:

- breathlessness
- a cough that does not go away
- wheezing
- a fever, with a temperature over 37.5°C.

You should also tell them if any existing breathing problems get worse. You may have:

- tests to check your lungs
- treatment, if needed.

Effects on the brain

Rarely, this treatment causes a brain condition that can be serious. You can make a full recovery from this. But it must be diagnosed and treated quickly.

This condition can cause:

- a headache that does not get better
- drowsiness or confusion
- changes in eyesight
- fits (seizures).

If you have any of these symptoms, it is important to either:

- contact the hospital straight away on the 24-hour number
- go to the hospital straight away.

You should not drive yourself to hospital.

Other important information

Second cancer

This treatment can increase the risk of developing a second cancer years later. This is rare. The benefits of treatment usually far outweigh this risk. Your doctor, nurse or pharmacist can give you more information.

Irradiated blood and platelets

After this treatment, if you need blood or platelet transfusions, they should be treated with radiation (irradiated) before you have them. This helps prevent a rare but serious reaction that can be caused by white blood cells in the donated blood.

Irradiated blood is not radioactive and will not harm you or anyone you come into contact with.

Some people must have any blood or platelet transfusions irradiated for the rest of their lives. If you need this, your doctor will give you more information. They will give you a medical alert card to carry in case you need blood in an emergency or at another hospital.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

Vaccinations

Cancer doctors usually recommend that people with cancer have vaccinations for flu and [coronavirus \(covid\)](#). They may also recommend other vaccines, such as Shingrix[®] for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they protect you against. There are several live vaccines, including the yellow fever vaccine.

It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them.

Alcohol

Some preparations of this treatment may contain alcohol. If having alcohol is a problem for you, tell your doctor, nurse or pharmacist. Your blood alcohol level may be above the legal limit after you have

the treatment. Do not drive or operate machinery for a few hours after having this treatment, even if you feel okay.

Contraception

Your doctor, nurse or pharmacist will advise you not to [get pregnant](#) or [make someone pregnant](#) while having this treatment and for some time afterwards. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

Breastfeeding

You are advised not to [breastfeed](#) while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Fertility

Some cancer drugs can affect whether you can [get pregnant](#) or [make someone pregnant](#). If you are worried about this, it is important to talk with your doctor before you start treatment.

Sex

It is possible that small amounts of chemotherapy may be passed on through vaginal fluids or semen. If you have sex in the first few days after treatment, your cancer team will usually advise using condoms or a dental dam to protect your partner.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

[Cancer Information Development team](#)

Author

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

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